

XSL - How to remove unused namespaces from source xml?

XSL - How to remove unused namespaces from source xml? - Author not stated, Stack Overflow.

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Content

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This question shows research effort; it is useful and clear

19

This question does not show any research effort; it is unclear or not useful

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I have an xml with a lot of unused namespaces, like this:

```
<?xml version="1.0" encoding="UTF-8"?>
<ns1:Envelope xmlns:ns1="http://www.a.com" xmlns:ns2="http://www.b.com"
xmlns:ns3="http://www.c.com" xmlns:ns4="http://www.d.com">
  <ns1:Body>
    <ns2:a>
      <ns2:b>data1</ns2:b>
      <ns2:c>data2</ns2:c>
    </ns2:a>
  </ns1:Body>
</ns1:Envelope>
```

I would like to remove the unused namespaces without having to specify in the xslt which ones to remove/maintain. The result xml should be this:

```
<?xml version="1.0" encoding="UTF-8"?>
<ns1:Envelope xmlns:ns1="http://www.a.com" xmlns:ns2="http://www.b.com">
  <ns1:Body>
    <ns2:a>
      <ns2:b>data1</ns2:b>
      <ns2:c>data2</ns2:c>
    </ns2:a>
  </ns1:Body>
</ns1:Envelope>
```

I've googled a lot but haven't found a solution to this particular issue. Is there any?

Thanks.

PS: Not 100% sure but I think it should be for XSL 1.0.

- [xml](#)
- [xslt](#)
- [xml-namespaces](#)

[edited Jan 4, 2011 at 12:18](#)

[asked Jan 4, 2011 at 11:59](#)

[

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Unlike the answer of @Martin-Honnen, this solution produces exactly the desired result -- the necessary namespace nodes remain where they are and are not moved down.

Also, this solution correctly deals with attributes that are in a namespace:

```

<xsl:stylesheet version="1.0"
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform">
  <xsl:output omit-xml-declaration="yes" indent="yes"/>
  <xsl:strip-space elements="*" />

  <xsl:template match="node()|@" priority="-2">
    <xsl:copy>
      <xsl:apply-templates select="node()|@" />
    </xsl:copy>
  </xsl:template>

  <xsl:template match="*">
    <xsl:element name="{name()}" namespace="{namespace-uri()}">
      <xsl:variable name="vtheElem" select="." />

      <xsl:for-each select="namespace::*">
        <xsl:variable name="vPrefix" select="name()" />

        <xsl:if test=
          "$vtheElem/descendant::*
            [(namespace-uri()=current()
              and
                substring-before(name(),':') = $vPrefix)
            or
              @*[substring-before(name(),':') = $vPrefix]
            ]"
          ">
          <xsl:copy-of select="." />
        </xsl:if>
      </xsl:for-each>
      <xsl:apply-templates select="node()|@" />
    </xsl:element>
  </xsl:template>
</xsl:stylesheet>

```

when this transformation is applied on the following XML document (the provided XML document with an added namespaced attribute):

```
<ns1:Envelope xmlns:ns1="http://www.a.com" xmlns:ns2="http://www.b.com"
xmlns:ns3="http://www.c.com" xmlns:ns4="http://www.d.com">
  <ns1:Body ns2:x="1">
    <ns2:a>
      <ns2:b>data1</ns2:b>
      <ns2:c>data2</ns2:c>
    </ns2:a>
  </ns1:Body>
</ns1:Envelope>
```

the desired, correct result is produced:

```
<ns1:Envelope xmlns:ns1="http://www.a.com" xmlns:ns2="http://www.b.com">
  <ns1:Body ns2:x="1">
    <ns2:a>
      <ns2:b>data1</ns2:b>
      <ns2:c>data2</ns2:c>
    </ns2:a>
  </ns1:Body>
</ns1:Envelope>
```

edited Nov 10, 2017 at 16:31

[

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answered Jan 4, 2011 at 14:24

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[user357812](#)

@mdiez: There is a problem with namespaces... Some implementations don't handle XPath namespace axe.

2011-01-04T17:51:05.08Z+00:00

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tyg

To expand on the comment of Implementations unaware of the namespace axis: This will result in namespaces being "pushed down" to each element that uses the namespace. This, f.e., applies to the default TransformerFactory of Java, whereas the Saxon implementation handles this correctly.

2017-11-10T15:50:05.063Z+00:00

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tyg

If you like to preserve namespaces that only occur in attribute *values* and therefore are not syntactically necessary, see @Gentil's answer below.

2017-11-10T16:07:26.663Z+00:00

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Dimitre Novatchev

@Leviathan, I wouldn't recommend trying to guess whether the string value of an attribute is a QName or just happens to be a syntactically-valid QName -- in the general case this is just guessing. One could use schema information, if it is known that the XML document is an instance of a given schema.

2017-11-10T16:32:37.077Z+00:00

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Dimitre Novatchev

@Xyaren This means that apache-xalan is buggy. The transformation in this answer is standard (no extensions) XSLT 1.0 and should produce the same results with any compliant XSLT 1.0 processor.

2021-03-10T15:46:02.3Z+00:00

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Well if you use

```
<xsl:stylesheet
  xmlns:xsl="http://www.w3.org/1999/XSL/Transform"
  version="1.0">

  <xsl:template match="@* | text() | comment() | processing-instruction()">
    <xsl:copy/>
  </xsl:template>

  <xsl:template match="*">
    <xsl:element name="{name()}" namespace="{namespace-uri()}">
      <xsl:apply-templates select="@* | node()" />
    </xsl:element>
  </xsl:template>

</xsl:stylesheet>
```

then unused namespaces are removed but the result is more likely to look like

```
<ns1:Envelope xmlns:ns1="http://www.a.com">
  <ns1:Body>
    <ns2:a xmlns:ns2="http://www.b.com">
      <ns2:b>data1</ns2:b>
      <ns2:c>data2</ns2:c>
    </ns2:a>
  </ns1:Body>
</ns1:Envelope>
```

than what you asked for.

answered Jan 4, 2011 at 12:49

[

[\[image: Martin Honnen's user avatar\]](#)

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[Martin Honnen](#)

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2 Comments

user357812

+1 Also good answer without the "not always implemented" `namespace` axe. But it could end up with a lot of "duplicated" namespace declarations for elements being the first under the namespace URI for each branch.

2011-01-04T17:55:15.083Z+00:00

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james.garriss

Perfectly good answer. It's semantically equivalent; the location of the namespace declarations doesn't matter.

2012-07-19T19:31:59.723Z+00:00

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Adding to Dimitre's answer, if those namespaces should be preserved that only occur in attribute *values*, add this condition: `@*[contains(.,concat($vPrefix,':'))]` :

```
<xsl:if test= "$vtheElem/descendant::* [namespace-uri() = current() and
    substring-before(name(),':') = $vPrefix or
    @*[substring-before(name(),':') = $vPrefix] or
    @*[contains(.,concat($vPrefix,':'))]
]">
```

This will correctly preserve the namespace `ns3` because of `attrib="ns3:Header"` as in the following example.

```
<ns1:Envelope xmlns:ns1="http://www.a.com" xmlns:ns2="http://www.b.com"
xmlns:ns3="http://www.c.com" xmlns:ns4="http://www.d.com">
  <ns1:Body ns2:x="1">
    <ns2:a>
      <ns2:b attrib="ns3:Header">data1</ns2:b>
      <ns2:c>data2</ns2:c>
    </ns2:a>
  </ns1:Body>
</ns1:Envelope>
```

edited Nov 11, 2017 at 0:03

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tyg

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answered Feb 9, 2017 at 16:35

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